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An Adjoint Technique for Estimation of Interstation Phase and Group Dispersion from Ambient Noise Cross Correlations

by Rhys Hawkins* and Malcolm Sambridge

Abstract A method of extracting group and phase velocity dispersions jointly for Love- and Rayleigh-wave observations is presented. This method uses a spectral element representation of a path average Earth model parameterized with density, shear-wave velocity, radial anisotropy, and V_P/V_S ratio. An initial dispersion curve is automatically estimated using a heuristic approach to prevent misidentification of the phase. A second step then more accurately fits the observed noise correlation function (NCF) between interstation pairs in the frequency domain. For good quality cross correlations with reasonable signal-to-noise ratio, we are able to very accurately fit the spectrum of NCFs and hence obtain reliable estimates of both phase and group velocity jointly for Love and Rayleigh surface waves. In addition, we also show how uncertainties can be estimated with linearized approximations from the Jacobians and subsequently used in tomographic inversions.

Introduction

Ambient noise tomography is now a well-established technique that uses the cross correlation of the ambient seismic-wave field, excited by ocean swells, storms, and wind, to estimate so-called empirical Green's functions (EGFs) between two seismic recording stations. From these EGFs, estimates of surface-wave dispersions can be obtained and can subsequently be used in tomographic inversions of the Earth's near-surface structure (Campillo and Paul, 2003; Shapiro and Campillo, 2004; Paul *et al.*, 2005; Sabra *et al.*, 2005).

Early work by Aki (1957) on microtremors pioneered using noise correlations to predict interstation surface-wave propagation properties. Subsequently, methods have been extended to estimating EGFs (Lobkis and Weaver, 2001; Derode *et al.*, 2003; Snieder, 2004; Wapenaar, 2004; Larose *et al.*, 2005) (see also Larose *et al.*, 2006, for a review article).

The estimation of EGFs from ambient noise cross correlations requires significant preprocessing. An early study attempting to standardize these steps was Bensen *et al.* (2007), in which the authors made several recommendations, the major components being: time-domain normalization to remove local seismicity, and spectral whitening to improve resolvability of low-energy parts of the spectrum.

Bensen *et al.* (2007) also discuss the extraction of phase velocity, suggesting at the time that there was no known suitable method. For this reason, the authors propose extraction of the group velocity using frequency–time analysis (FTAN) techniques (Dziewonski and Hales, 1972; Herrmann, 2013). FTAN in turn became the method of choice for many ambient noise studies because of the existence of established codes and its ease of application.

Although group velocity information can be used for tomographic studies, arguably phase velocity is more useful. For instance, (1) the group velocity can be uniquely determined from phase velocity dispersion through differentiation. Conversely, estimating phase velocity from group velocity requires an unknown constant of integration. (2) The process of extracting group velocity from EGFs is imprecise due to the width of band-pass filtering being a trade-off between temporal and frequency resolution. This trade-off tends to make observed envelopes wider, hence less precise, and also admits a higher possibility of interference from other phases (Boschi *et al.*, 2013). (3) The FTAN approach uses a far-field approximation because it was originally intended for earthquake-driven surface-wave studies. This approximation means that the frequency range is limited for nearby stations with the commonly used rule of thumb stating that observations from station pairs within three wavelengths should be discarded as unreliable.

Motivated by the benefits of phase velocity over group velocity dispersion information, several methods have been proposed for the extraction of phase velocity dispersion from EGFs. The first is an image-based time-domain technique (Yao *et al.*, 2005, 2006) that similarly relies on a far-field approximation. Frequency-domain techniques (Ekström *et al.*, 2009; Ekström, 2014; Menke and Jin, 2015) based on an analysis of the statistics of microtremor correlations between stations (Aki, 1957) do not use the far-field

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approximation and therefore can extract longer period dispersion information from closer station pairs.

Forming the basis for a number of frequency-domain techniques is the result from Aki (1957), who showed the real component of correlated noise between stations is of the form of a Bessel function of the first kind with zeroth order. Restating this result here,

$$\bar{\rho}(\omega_0, r) = J_0\left(\frac{\omega_0}{c(\omega_0)} r\right), \tag{1}$$

in which $\bar{\rho}$ is the real component of the Fourier transform of the noise correlation function (NCF), ω_0 is the angular frequency of the fundamental mode, $c(\omega_0)$ is the frequency-dependent phase velocity, and r is the interstation distance. The original Ekström *et al.* (2009) approach uses the zero crossings of the real component of the spectrum of the NCF to construct phase velocity curves directly. From the observed zero crossings in the real component of the Fourier-transformed NCFs (i.e., a set of zero crossings $z_1 \ldots z_n$ at angular frequencies $\omega_1 \ldots \omega_n$), trial phase velocity curves are constructed using

$$c_m(\omega_n) = \frac{\omega_n r}{z_{n+2m}}, \tag{2}$$

with m as the integer trial value that is 0, ± 1, ± 2, and so on. From these trial curves, the dispersion curve most closely matching a reference dispersion curve for the region is chosen as the observed phase velocity curve. A potential problem of this approach is that noise inherent in the observations can cause spurious zero crossings, as highlighted by Menke and Jin (2015). Misidentification of missed or spurious zero crossings can result in phase velocity dispersion curves that drop precipitously to unfeasible velocities. These problems were recognized, and a subsequent improvement to the method was introduced by Ekström (2014) that adds the extra step of fitting a spline to the real component of the spectrum aimed at eliminating spurious zero crossings. A further extension of this general approach was the inclusion of completely fitting the real part of the spectrum by Menke and Jin (2015), which subsequently improved the rejection of spurious zero crossings and added the ability to use residuals from the inversion of individual cross-correlated station pairs as quality factors.

In this study, we aim to build on these advances by developing a nonlinear optimization approach for extraction of phase velocity information. A key factor in this new approach is the use of a path average Earth model as a constraint that allows us to invert for Love and Rayleigh dispersion curves in a single inversion and also to estimate uncertainties using standard methods of nonlinear inverse theory.

Method

The goal of the inversion method presented here is to estimate a surface-wave dispersion curve from the observed real component of the spectra of the NCF. In almost all geophysical inversion, the observations do not uniquely determine a solution, and some form of prior information or constraints are required. This is apparent in previous work, for example Ekström *et al.* (2009), in which the approach taken was to compute several potential dispersion curves constructed from the observed zero crossings and choose one that was within an *a priori* determined range of phase velocities. Recent alternatives to this method by Menke and Jin (2015) similarly include a first stage grid search constrained between plausible phase velocities, followed by a second stage refinement that incorporates penalties on deviation from the first stage phase velocity and smoothness.

In our approach, we propose using a 1D Earth model as our prior on plausible phase velocities. The structure of the Earth determines the observed phase velocity between two points of the Earth. Under the assumption that lateral heterogeneity is slowly varying, a 1D Earth model can be used to represent the path average from which dispersion curves can be calculated. At lower frequencies, this is a reasonable approximation because abrupt lateral changes in structure are less well sensed by long-wavelength surface waves. At high frequencies, this assumption may not hold because scattering and mode conversions are more likely. However, mode conversions are generally dominated by higher order modes converting to the fundamental and likely have little influence in ambient noise studies in which higher order mode excitation is small (Datta *et al.*, 2017).

Across much of the Earth, we have a good deal of prior information on the likely structure that can be used as additional constraint. Examples include global reference models (Dziewonski and Anderson, 1981; Kennett *et al.*, 1995), globally constrained crustal models (Laske *et al.*, 2013), regionalized upper-mantle models (Gudmundsson and Sambridge, 1998), or information on the class of crust such as continental, spreading ridge, or oceanic (Knopoff, 1972; see his fig. 11). Given such prior information, our approach constitutes a standard geophysical inverse problem in which a 1D Earth model, constrained by prior information, is used to predict the spectra of NCFs.

Currently, most ambient noise tomographic inversions use a two-step process: first, the NCF is used to determine group or phase velocity dispersion between individual seismic station pairs; and second, group or phase velocities from a collection of seismic station pairs are used to determine Earth structure. We propose estimation of both group and phase velocity and a mean Earth structure in a single step through fitting the observed real component of the Fourier transformed NCF $\rho^{\text{obs}}(\omega, r)$. Because of the oscillatory nature of $\rho^{\text{obs}}(\omega, r)$, we need to include an initial heuristic estimate of the phase velocity in an attempt to avoid local minima in fitting, so our proposed method overall consists of the following steps:

1. preprocessing to obtain NCFs and EGFs (although not a main feature of this study, a simple approach is described herein for completeness);

2. heuristic estimation of phase velocity dispersion curves from zero crossings and peaks and troughs of the spectrum of the NCFs, followed by optimization of a path average Earth model to fit these preliminary dispersion curves;
3. nonlinear optimization of a path average Earth model to fit the real part of the spectrum of the NCFs; and
4. estimation of uncertainties for the dispersion curves using linear approximations of Jacobians obtained from the inversions.

These steps are detailed in the following sections.

Preprocessing

We briefly describe our strategy for obtaining NCFs and EGFs. This consists of the following steps: day-length records are obtained for three-component records and instrument responses are removed. Filtering is performed using a linear phase filter between 10 mHz and 1 Hz with signals resampled to 2 Hz. Trends are removed and each day-length signal is normalized so that it is approximately distributed according to $N(0,1)$. This time-domain normalization is used in preference to the more commonly used 1 bit normalization, or time-domain weighted averaging, because these have nonlinear effects on the phase of the signals. The day-length signals are then decomposed into overlapping hour-long windows to improve stacking (Seats *et al.*, 2012) and Fourier transformed to the frequency domain. The influence of seismicity on the ambient noise record can be reduced by removing windows with signals above an amplitude threshold. Cross correlation and stacking are performed in the frequency domain incorporating amplitude normalization, which fills the role of spectral whitening as in Ekström (2014). The NCF obtained directly from the stacked cross correlations is stored as a function of frequency and the EGF is obtained by differentiating the NCF.

Quality checks are performed by band-pass filtering at 0.1 Hz and verifying that a consistent signal appears on both causal and acausal sides of the EGFs in the time domain (i.e., both the envelope and signal appear to be coherent, suggesting a fundamental surface wave is present). Signal-to-noise ratios are computed in the 0.1 Hz band from the ratio of the maximum amplitude in a window in time corresponding to 3 ± 1 km/s derived from the interstation distance, and times after 5 km/s for both causal and acausal branches.

It should be noted here that our preprocessing method is relatively simplistic and more advanced approaches are now common practice (Schimmel and Paulssen, 1997). Advances such as these could be used to further improve the recovery of phase velocity.

Initial Phase Velocity Optimization

As mentioned earlier, we use a prior model appropriate for the region of study. It is possible that this model may predict phase velocity dispersion that induces cycle skipping in the fitting of the real component of the NCF, due to the oscillatory nature of Bessel functions. Because the oscillatory nature of the real part of the NCF is predominantly determined by the interstation distance, with larger distances corresponding to higher frequency oscillations, we choose to prefit a phase velocity dispersion curve generated in a similar manner to earlier studies (Ekström *et al.*, 2009). In addition to using zero crossings of the Bessel function, we additionally fit peaks and troughs.

Given equation (1) and assuming slowly varying amplitude with frequency, the peaks and troughs are defined by the zeros of the gradient of the Bessel function

$$\frac{\mathrm{d}}{\mathrm{d}\omega_0}\bar{\rho}(\omega_0, r) \propto -J_1\left(\frac{\omega_0}{c(\omega_0)}r\right), \tag{3}$$

in which J_1 is the first-order Bessel function of the first kind, for which we can precompute additional zeros that provide additional points on the phase dispersion curve. We then use an iterative heuristic approach to determine the correct phase dispersion curve with the following steps:

1. Select the largest magnitude peak or trough of the NCF as a starting point.
2. Select the zero of the first-order Bessel function that minimizes the difference between the prior model dispersion curve at the largest peak or trough frequency using the peak or trough equivalent of Ekström *et al.* (2009, their equation 3). This defines the integer offset to start from for forming a curve, that is, m in equation (2).
3. Proceed from the starting peak or trough in positive and negative frequency directions by using the reference model and the current point in frequency and phase velocity to predict a window for the next zero crossing, peak, or trough.
4. For proposed peaks or troughs: if the amplitude is smaller than 5% of the maximum, reject the peak or trough. This is intended to prevent misidentification of peaks or troughs when there are low-amplitude signals.
5. For proposed zero crossings: using the gradient of the prior model's dispersion curve as a guide, if the inclusion of the next zero crossing would cause a sudden jump in the dispersion curve, reject this zero crossing. This rejection criterion is used to skip spurious zero crossings causing sudden jumps in dispersion curves.
6. If a peak, trough, or zero crossing is not found or rejected, continue searching by predicting the next window for a peak, trough, or zero crossing.
7. Once the desired frequency range of the NCF has been scanned, use the lowest frequency trough found to estimate the predicted frequency of the first trough of the Bessel function. Use this to adjust the integer offset of the selected zeros by ± 2 if the predicted first trough is in negative frequency or too large. This assumes that the prior model produces a phase velocity pick that is at most incorrect by a phase of 2π.

The key parts of the algorithm outlined previously are designed to prevent two things: first, the rejection of peaks, troughs, or zero crossing based on a search window (item 3) and rejection based on a gradient deviation criteria (item 5) to prevent spurious extra peaks, troughs, or zero crossings that can cause dispersion curve jumps. Second, the algorithm also keeps track of the zero of either the first- or second-order Bessel functions to ensure a consistent dispersion is obtained. The benefit of including peaks and troughs means there are generally more points along the dispersion curve. In addition, peaks and troughs are more easily identified and less affected by noise than the zero crossings.

Uncertainties are estimated based on the half-width between the surrounding dispersion curves; that is, the uncertainty is set to half the distance to the next curve. This can easily be calculated by using the zeros of the first- and second-order Bessel functions in combination with a ± 2 offset from the determined best dispersion curve.

Dispersion curves obtained in a similar manner were used directly by Ekström *et al.* (2009). In our method, we take the additional step of using these estimated dispersion curves in an inverse problem to preoptimize a path average Earth model. Given an observed phase velocity dispersion function $c^{\mathrm{obs}}(\omega)$ computed using the heuristic procedure mentioned earlier, and a predicted phase velocity dispersion $c^{\mathrm{pred}}(\omega, \mathbf{m})$ from an Earth model $\mathbf{m}$, we can minimize

$$\sum_i \frac{(c^{\mathrm{pred}}(\omega_i, \mathbf{m}) - c^{\mathrm{obs}}(\omega_i))^2}{\sigma^{\mathrm{obs}}(\omega_i)} + (\mathbf{m}-\mathbf{m_0})^T C_m^{-1} (\mathbf{m}-\mathbf{m_0}), \quad (4)$$

in which $\sigma^{\mathrm{obs}}(\omega_i)$ is the estimated uncertainties at each discrete frequency ω_i of interest, C_m is the model covariance matrix, and $\mathbf{m_0}$ is the prior or reference Earth model.

The 1D Earth model parameterization consists of spectral elements, and a gradient descent approach is used to minimize equation (4). Here, the gradient of the misfit (equation 4) is calculated with the adjoint approach described by Hawkins (2018b). The purpose of this initial fitting is to generate a starting model from which to perform the final nonlinear optimization.

Nonlinear Fitting of the NCF

The next step is fitting the real component of the Fourier transformed NCF; that is

$$\rho^{\mathrm{obs}}(\omega, r) = \mathbb{R}(\mathcal{F}N(t)), \quad (5)$$

in which $N(t)$ is the NCF in the time domain, $\mathcal{F}$ is the Fourier transform (which transforms from time t to angular frequency ω), and $\mathbb{R}$ signifies the real part.

From a given path average Earth model, we can predict the Bessel function representing the real component of the noise spectrum based on the phase velocity dispersion curve; that is

$$\rho^{\mathrm{pred}}(\omega, r, \mathbf{m}) = J_0\left(\frac{\omega r}{c(\omega, \mathbf{m})}\right), \quad (6)$$

in which J_0 is the zero-order Bessel function of the first kind (Aki, 1957). The goal of our inversion (and many others) is to find an Earth model for which the difference between $\rho^{\mathrm{obs}}(\omega, r)$ and $\rho^{\mathrm{pred}}(\omega, r, \mathbf{m})$ is minimized. However, a commonly encountered problem is that, because of the unevenly distributed noise sources and their varying amplitude with frequency, the amplitude of $\rho^{\mathrm{obs}}(\omega, r)$ is not the same as predicted by the Bessel function, even though its zero crossings occur in the same locations. This mismatch presents a problem because an inversion will tend to find a compromise solution that fits both amplitude and phase (and possibly both poorly), whereas we would prefer an inversion that ignores amplitude mismatch and preferentially fits phase because most of the structural information is contained in the phase.

Our solution is to effectively correct $\rho^{\mathrm{obs}}(\omega, r)$ so that its amplitude always matches that of the data using the analytical signal representation (Bozdağ *et al.*, 2011). We begin by noting that an arbitrary function $f(\omega)$ can be represented as

$$f(\omega) = A(\omega) \exp i\phi(\omega), \quad (7)$$

in which $A(\omega)$ is its envelope, and $\phi(\omega)$ is its instantaneous phase. The envelope can be computed using the Hilbert transform; that is

$$A(\omega) = \mathcal{E}(f(\omega)) = |f(\omega) + i\mathcal{H}(f(\omega))|, \quad (8)$$

in which $\mathcal{E}(f(\omega))$ is the envelope of the arbitrary function $f(\omega)$, and $\mathcal{H}$ is the Hilbert transform (acting on independent variable ω in this case). We can then define an amplitude corrected prediction as

$$\rho^{\mathrm{corr}}(\omega, r, \mathbf{m}) = \frac{\mathcal{E}(\rho^{\mathrm{obs}}(\omega))}{\mathcal{E}(\rho^{\mathrm{pred}}(\omega, r, \mathbf{m}))} \rho^{\mathrm{pred}}(\omega, r, \mathbf{m}), \quad (9)$$

and perform an inversion based on minimization of the error between $\rho^{\mathrm{obs}}(\omega)$ and $\rho^{\mathrm{corr}}(\omega)$.

A useful analytical result is that the envelope of a zero-order Bessel function of the first kind is given by

$$\mathcal{E}(J_0(x)) = \sqrt{J_0(x)^2 + Y_0(x)^2}, \quad (10)$$

in which Y_0 is the zero-order Bessel function of the second kind, from which we can rewrite equation (9) as

$$\rho^{\mathrm{corr}}(\omega, r, \mathbf{m}) = \frac{\mathcal{E}(\rho^{\mathrm{obs}}(\omega))}{\sqrt{J_0(k(\omega, \mathbf{m})r)^2 + Y_0(k(\omega, \mathbf{m})r)^2}} J_0(k(\omega, \mathbf{m})r), \quad (11)$$

which now represents the forward model as a function of $k(\omega, \mathbf{m})$ or the wavenumber. An analytic partial derivative of equation (11) is possible

$$\frac{\partial \rho^{\mathrm{corr}}}{\partial k} = \frac{\mathcal{E}(\rho^{\mathrm{obs}}(\omega))\left\{-rJ_1(rk) - rJ_0(rk)\frac{J_0(rk)J_1(rk)+Y_0(rk)Y_1(rk)}{J_0(rk)^2+Y_0(rk)^2}\right\}}{\sqrt{J_0(rk)^2+Y_0(rk)^2}} \tag{12}$$

with $k(\omega, \mathbf{m})$ abbreviated to k.

Given a starting model from the previous section parameterized in terms of varying shear-wave velocity, density, radial anisotropy, and V_P/V_S ratio with depth, we can use the adjoint technique of Hawkins (2018b) in a nonlinear gradient optimization to minimize

$$\sum_i \frac{(\rho^{\mathrm{corr}}(\omega_i, r, \mathbf{m}) - \rho^{\mathrm{obs}}(\omega_i))^2}{\sigma_i^2} + (\mathbf{m} - \mathbf{m_0})^T C_m^{-1} (\mathbf{m} - \mathbf{m_0}), \tag{13}$$

in which σ is an error level estimated from quiescent parts of the spectrum. Given an expected degree of cross talk between neighboring frequency bins in a spectrum, the independent and identically distributed noise model used previously is a simplifying assumption, and investigating covariance in this step may be a future extension. A prior on the deviation from a reference model is again used; in this case, however, $\mathbf{m_0}$ is the model obtained from the fitting of the initial phase velocity estimate.

Uncertainties

Uncertainties on the model parameters can be estimated from a linearity approximation of the posterior covariance matrix (Tarantola, 2005)

$$\widetilde{\mathbf{C}}_M \approx (\mathbf{G}^t \mathbf{C}_D^{-1} \mathbf{G} + \mathbf{C}_M^{-1})^{-1}, \tag{14}$$

in which the matrix $\mathbf{G}$ is the matrix of partial derivatives at the optimal model $\widetilde{\mathbf{m}}$ from the previous minimization. This gives an estimate of uncertainty of the model parameters; however, in many cases the observation desired is an estimate of phase or group velocity at a particular frequency for subsequent tomographic inversion. Similar to the matrix $\mathbf{G}$, matrices of partial derivatives of phase or group velocity with respect to model parameters can be computed at the model $\widetilde{\mathbf{m}}$, and an estimate of the covariance matrix for phase velocity for example can be computed with

$$\widetilde{\mathbf{C}}_{\mathrm{phase}} = \mathbf{J}_{\mathrm{phase}}^T \widetilde{\mathbf{C}}_M \mathbf{J}_{\mathrm{phase}}, \tag{15}$$

in which $\mathbf{J}_{\mathrm{phase}}$ is the Jacobian matrix of partial derivatives $\frac{\partial c(\omega_i)}{\partial m_j}$.

Results

We applied this technique to a collection of ambient noise cross correlations from Iceland. We use the same set of stations as Gudmundsson *et al.* (2007) that correspond to the Iceland Hotspot project temporary stations (see Data and Resources) and the BORG permanent station. In total, we have 459 station pairs that have at least one month of recording, with station geometry shown in Figure 1.

Prephase

Here, we show the first step in the estimation process in which an initial phase velocity dispersion curve is estimated using the peaks, troughs, and zero crossings of the real component of the NCF. For the prior model, we use a combined prior derived from two regional studies of Iceland (Li and Detrick, 2006; Green *et al.*, 2017), and use independent standard deviations of 0.5 gcm^{-3}, 0.5 kms^{-1}, 0.05, and 0.10 for density, shear-wave velocity, radial anisotropy, and V_P/V_S ratio, respectively.

In Figure 2a,b, we show the collection of all data estimated with our algorithm for both Love and Rayleigh data. This step of the method is quite quick, being of the order of 10 min for all 459 data running serially on a desktop computer.

From the plots, the phase velocity estimates are generally well grouped with approximately 13 station pairs in which the phase has likely been incorrectly identified, suggesting that the algorithm has generally done a good job of correctly identifying the phase (97% success rate). Our approach performed successfully over a large range of interstation distances, from approximately 40 to 300 km, and is run automatically without manual intervention. It is also clear that the spread of the Love-wave data is qualitatively larger, and we suspect this is due to its often higher level of noise or cross talk contamination of Love-wave observations over that of Rayleigh.

Phase

In the next step, the path average Earth model is optimized to predict the real component of the NCF. Overall, this process produces smoother versions of the dispersion curves than the noisy heuristic estimates. This can be seen in the summary of all station pairs in Figure 2 by comparing Figure 2a and b to Figure 2c and d. This process takes on the order of 5 min on a desktop computer for each station pair.

For a more detailed account of the results, we first show the results for a qualitatively good station pair. In Figure 3, the results for the processing of interstation pair HOT05 and HOT25 are shown. In Figure 3a, we show the final estimated curve for Love and Rayleigh dispersions with black dashed (Love) and solid (Rayleigh) lines. The dispersion curve for the reference model is shown with faint gray lines. This particular path crosses Iceland longitudinally and therefore across the spreading ridge. It appears slightly faster than the reference model at long periods and slower at short; however, the shortest period used for the construction of the reference model (obtained from results in Green *et al.*, 2017) was 6 s. Although this suggests an even slower shallow crust than previously observed, care should be taken in

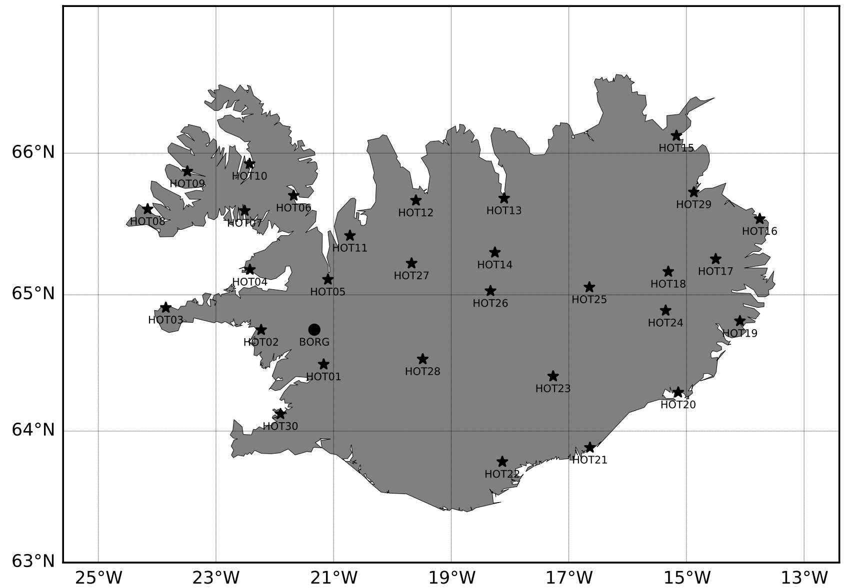

Figure 1. Collection of stations from the HOTSPOT deployment and one permanent station used.

the interpretation of high frequencies in ambient noise cross correlations because greater heterogeneity in the near-surface crust can introduce mode coupling and other multipathing or scattering effects.

In Figure 3b,c, we show the fits to the Bessel functions with the observed spectrum shown in gray and the recovered function in red. The fits are excellent for this station pair with only a small discrepancy in the Love-wave result around 0.30 Hz. Finally, in Figure 3d we show the path average models obtained for the four parameters in solid curves as functions of depth, with the reference model shown with dashed lines. The perturbations to the model are generally restricted to the near surface. In this formulation, layering is fixed thus generating artifacts near layer boundaries that could potentially be improved by allowing the layer thicknesses to vary as part of the inversion.

In Figure 4, we show another interstation pair that this time has a spectral hole in which the fitting is not as convincing. In Figure 4b, we can see that in the results for Love dispersion between 0.15 and 0.20 Hz, there is some disagreement between the fitted (red) and observed (gray) lines. In Figure 4c, for Rayleigh dispersion the fit in the same region is exemplary. Here, the benefit of fitting jointly Love and Rayleigh dispersion is apparent because having both sources of information allows, to some degree, inference of the dispersion in the presence of poor signal. In this example, there is good signal in the Rayleigh-wave NCF between 0.15 and 0.20 Hz, and this is fitted well and provides additional constraint when the Love-wave observations are poor.

Group Velocity

Although we do not explicitly fit group velocities, nonetheless group velocity dispersion can be computed because a path average Earth model is available. In Figure 5, we show the inversion results for the two demonstration paths for Love- and Rayleigh-group velocity. We also show as shaded

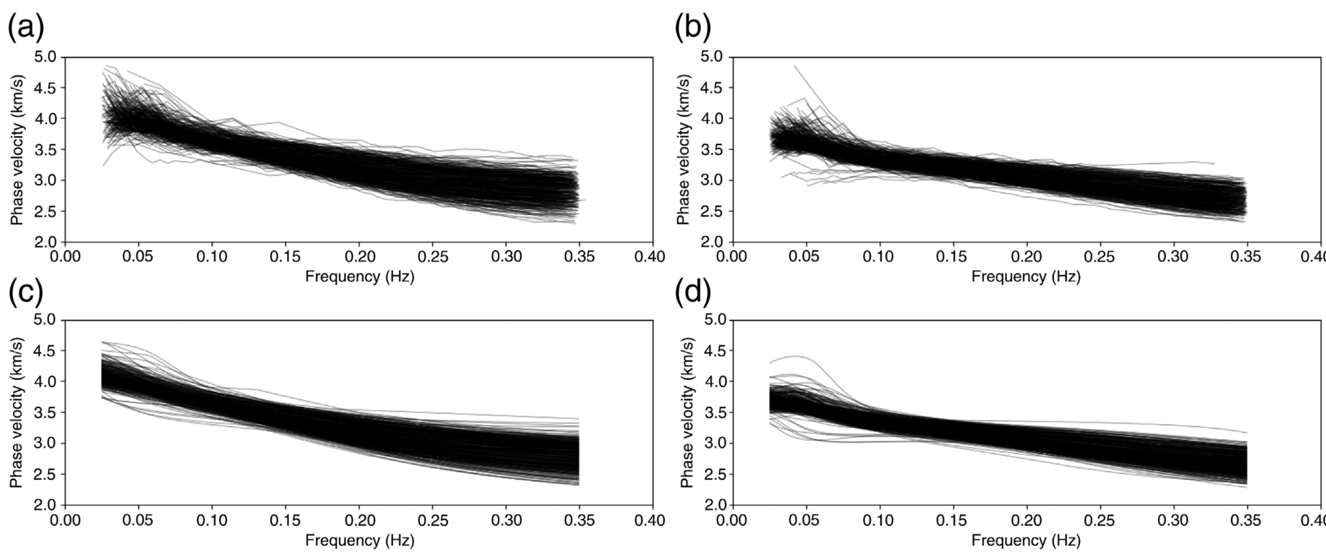

Figure 2. All station pair plots of phase velocity dispersion estimated using the initial heuristic approach for (a) Love-wave data, (b) Rayleigh-wave data, and the final result after nonlinear inversion in (c,d) for Love- and Rayleigh-wave data, respectively. These estimates are well grouped with few outliers.

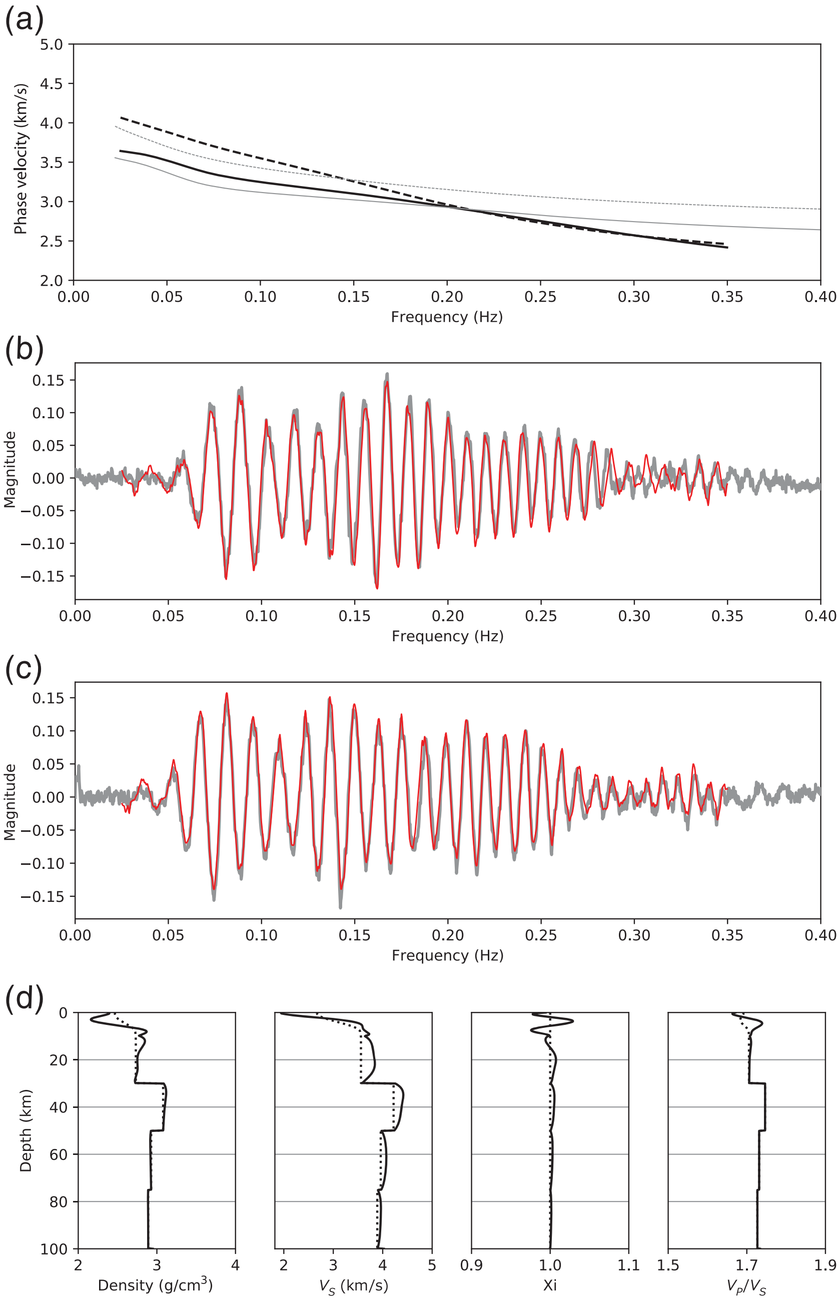

Figure 3. For HOT05–HOT25 station pair, the recovered phase velocities are shown in (a) for Love (dashed) and Rayleigh (solid) with faint lines showing the reference phase velocities. Fits of the Bessel functions are shown in (b) for Love-wave data and (c) for Rayleigh-wave data with observed in gray and fitted curve in red. (d) The parameterized path average Earth model where the final optimal model is shown with solid lines and the reference with dashed lines.

the signal energy in the acausal branch of the cross correlation obtained through FTAN (Dziewonski and Hales, 1972) (here the horizontal axis is frequency rather than the more common period).

For each of the results, there is good agreement between the observed FTAN image and the predicted group dispersion curves. In Figure 5a, the high-frequency Love-group velocity dispersion has not been well fit; however, there is little signal at such high frequencies, as seen in Figure 3b. In Figure 5c, the cause of the poor fit of the Bessel function seen in Figure 4b can be identified as a spectral hole in the FTAN plot. A benefit of using a path average Earth model and jointly inverting Love- and Rayleigh-wave data is that when such spectral holes appear, some inference can still be made as to the group velocities but with slightly less confidence, as we will show in the next section.

Uncertainties

Through linear approximations, we can estimate uncertainties for both phase and group dispersion. In Figure 6, we show the phase velocity predictions and uncertainties for the two demonstration paths. The general pattern of uncertainties is a large broadening at low frequencies. There is a minor broadening apparent in some of the results at high frequencies. This is to be expected and is representative of the bandlimited nature of ambient noise cross correlations in that the dominant energy is generally in the 5–15 s period range (Bensen *et al.*, 2007).

An interesting feature in Figure 6c is that in the range 0.15–0.20 Hz, there is a small broadening in uncertainties associated with the spectral hole observed in the phase and group velocity results (shown zoomed in the figure). This suggests that the method is able to capture uncertainties due to spectral holes within ambient noise cross correlations. In the range of frequencies of interest, the uncertainties are quite small and may be unrealistically so, as is commonly the case in linearity estimates of uncertainty in nonlinear optimization problems. Nonetheless, these uncertainty estimates may be useful in providing relative weightings in a subsequent tomographic inversion of either phase or group velocity maps.

Using the Adjoint Dispersion Curves for Tomographic Inversion

From the method presented, we have continuous dispersions curves with uncertainties for Love and Rayleigh surface waves. To validate the retrieved dispersion information and its uncertainties, we performed inversions at 6 s period to

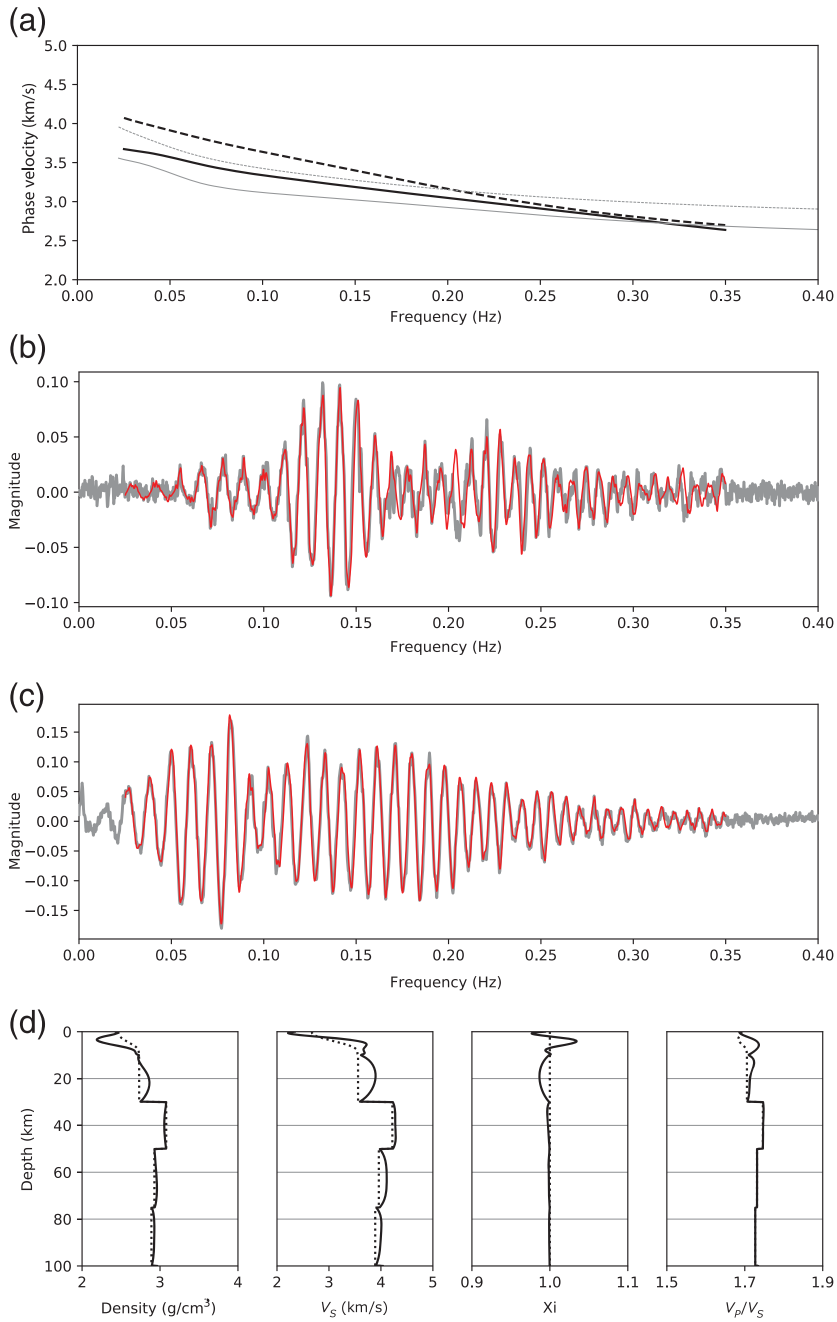

Figure 4. We show the same plots as in Figure 3 but for HOT05–HOT15 station pair. For this station pair, the fit for Love-wave data is poor between 0.15 and 0.20 Hz in (b).

compare previous studies by Gudmundsson *et al.* (2007) and Green *et al.* (2017).

Station pairs that had clearly observable phase mispicks were removed, as indicated earlier, and 2D tomographic maps of phase and group velocity were inverted for using the Bayesian trans-dimensional tree approach with a wavelet parameterization (Hawkins and Sambridge, 2015). This approach generates an ensemble of plausible models from which we can extract an ensemble mean as a single model for inspection. It also includes hierarchical sampling of the prior and hierarchical error estimation of a scaling factor to account for errors in the estimated uncertainties (Hawkins *et al.*, 2019). Inversions were simulated using six Markov chains with parallel tempering (Sambridge, 2014) for three million iterations, with 1.5 million iterations removed as burn-in samples to ensure adequate convergence that was qualitatively judged using the hierarchical error estimation as a proxy (Hawkins *et al.*, 2018).

The ensemble means for the four separate inversions are shown in Figure 7. From the plots, we can see that both Love and Rayleigh results are broadly consistent with slow velocity anomalies concentrated about the central part of Iceland, where the spreading ridge and volcanism are concentrated, and is in broad agreement with results from body-wave studies (Brandsdottir and Menke, 2008). In this study, we use the same source data from 31 stations as Gudmundsson *et al.* (2007), whereas Green *et al.* (2017) included data from more recent deployments for a total of 232 stations. The results presented here for the 6 s period of Rayleigh-group velocity inversion are in good agreement with the study of Green *et al.* (2017) with far less data, 444 ray paths compared with 2180. In this particular application, our improved method achieved a similar tomographic result with a factor of seven times fewer seismic stations.

In a validation of the linearized estimates of uncertainties, we can examine the posterior information on the hierarchical error scaling parameter used in the inversion. In a perfect inversion with perfect estimation of data errors, that is, our errors on phase and group dispersion were perfectly estimated, the hierarchical scaling parameter should obtain a posterior mode of one. The hierarchical error scaling attempts to account for errors in the estimation of data errors and in modeling approximations associated with the tomographic inversion. The history of hierarchical error scaling parameter and posterior histograms is shown in Figure 8. The relatively flat history in each of the plots shows good convergence and the hierarchical scaling parameter settled on between 0.4 and 1.8. This indicates that the linearized estimates of uncertainties provide reasonable estimates of the uncertainties. The tomographic reconstructions show similar features to studies with larger numbers of data (Green *et al.*, 2017), suggesting that these uncertainties

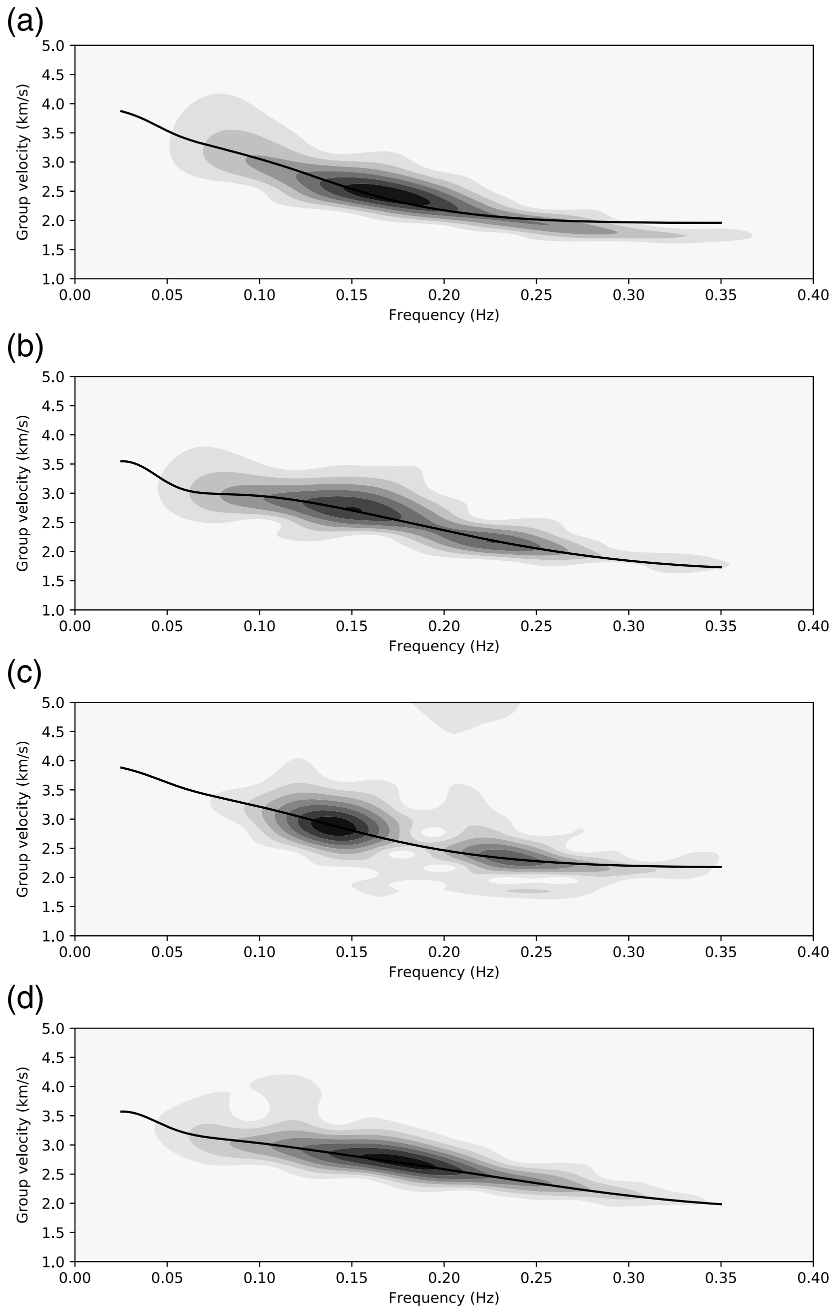

Figure 5. Group velocity predictions are shown for the two demonstration interstation paths with the frequency–time analysis (FTAN) plotted in the background. (a,b) Love- and Rayleigh-group velocity for the HOT05–HOT25 path where the fit is good, except for at high frequency for Love waves. (c,d) Results are shown for the HOT05–HOT15 path where the predictions are good. Of note is the fact that the group velocity is reliably predicted even in the presence of a spectral hole between 0.15 and 0.20 Hz in (c).

provide additional information that improves the recovered features.

In summary, we have shown in detail the recovery of phase and group velocity dispersion information for both Love- and Rayleigh-surface waves from ambient noise cross correlations using an adjoint-based optimization procedure. From the automated application of the method to 459 station pairs obtained from the Iceland study region, 13 station pairs appeared to be poorly estimated resulting in a 97% recovery rate, which is far higher than typical manual estimates of group velocity using FTAN. For example, in Hawkins (2018a), results from traditional FTAN on the same Icelandic dataset were reported to vary from 35% to 68% depending on the period (mean of 55% across all periods considered).

The key features of this approach is that it can be run without manual intervention and also produces continuous dispersion estimates with uncertainties that, when incorporated into tomographic inversions, produce results consistent with previous studies with significantly larger number of observations.

Discussion

We developed a method for inferring dispersion from ambient noise cross correlations using a nonlinear optimization of a path average Earth model. The benefits of this approach are that both phase and group velocities are recoverable for both Love and Rayleigh waves, their uncertainties can be estimated, and the process is largely automated (a final quality control pass may be required to remove a small number of misidentified phases).

This approach makes four key simplifying assumptions: first, only the fundamental mode is assumed to be propagating. For ambient noise studies, this is a reasonable approximation because there is rarely sufficient energy in higher order modes. For higher frequency studies for closely located arrays, this may not be the case. Second, the ambient noise distribution is sufficiently smooth so that errors in apparent arrival times between receivers due to the uneven distribution of noise sources are small (Weaver *et al.*, 2009). An extension to this approach would be to include noise source and Earth structure inversions together (e.g., Fichtner *et al.*, 2016). Third, the path average model does not take into account potential mode conversions between Love and Rayleigh at sharp velocity contrasts. This is more likely to be an issue at higher frequencies in which the propagation of Love and Rayleigh waves is more affected by the heterogeneous shallow crust. Finally, we used a simple

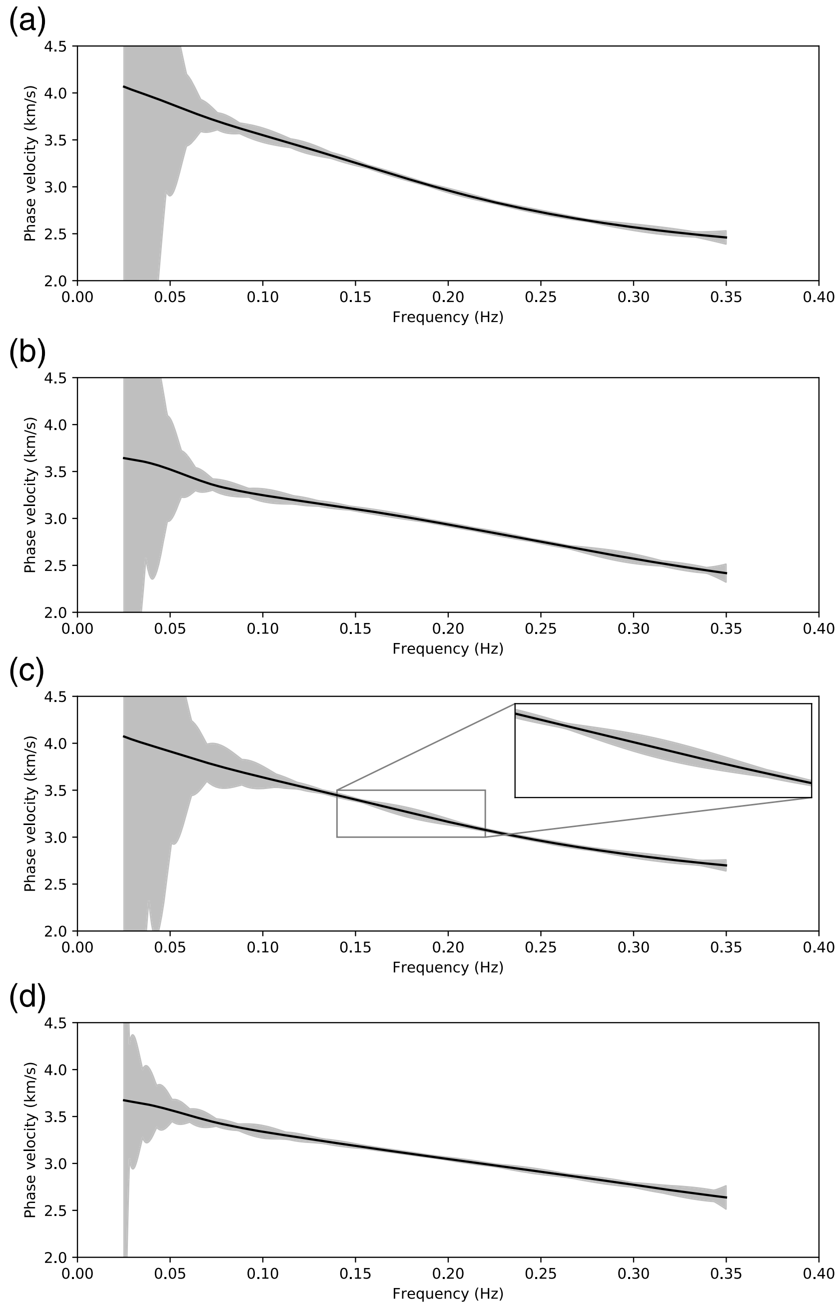

Figure 6. Through linearized estimates, we can compute 95% confidence intervals for both phase and group velocity predictions. The phase velocity uncertainties for Love and Rayleigh data in (a,b) for the HOT05–HOT25 station pair and (c,d) for the HOT05–HOT15 station pair. The general behavior is for a large broadening of uncertainties at low frequency with a corresponding but smaller broadening at higher frequencies. Of note is the presence of slightly higher uncertainties between 0.15 and 0.20 Hz in (c) corresponding to the spectral hole shown in Figure 5c.

independent Gaussian noise model for the errors on the observed NCF. This noise model may be too simple for some applications, and a correlated noise model would be more appropriate. Estimating correlated (and potentially nonstationary) noise covariances from NCFs would be an area of future work.

Another criticism may be that the use of a prior Earth model begs the question, what if the prior model has significant deviation from the truth? We argue that most of the deviation between a prior Earth model and a new regional study will be confined to the near surface or higher frequency in the NCF. As long as reasonable estimates are available for the lower crust, for example, from crustal models (Laske *et al.*, 2013), and the upper-mantle structure can be constrained, we expect proper identification of phase with this approach.

Perhaps the greatest weakness of the approach, as in many methods for identifying the phase, is in the heuristic approach of the identification of the phase. This component of the method could be improved further in the future. Because the method is automated, a potential area of future work could examine incorporating the tomographic inversion so that a 3D model of the region is used to predict the real component of all the available NCFs. In doing so, coherency between station pairs that have similar interstation paths can be taken into account, potentially improving recovery rates further. Also, the tomography results presented did not use the path average Earth models obtained from the nonlinear optimization, and there remains the possibility of combining these directly in an inversion for a 3D Earth model using all NCFs concurrently.

The 97% recovery rate found with the new algorithm represents an improvement over traditional manual FTAN approach in which a previous analysis of the same dataset recovered 35%–68% of station pairs depending on period. We hope it may be of assistance in improving ambient noise tomography studies across the globe. Of particular note is the fact that our tomographic results for Iceland are very similar to recent studies (Green *et al.*, 2017) that used a far larger number of stations (214 compared with 31 in this study). In addition, given the method's applicability to both Love and Rayleigh waves, this method has applications in studies of radial anisotropy of the crust. We also point out that the method could equally be adapted to invert only for Love- and Rayleigh-wave data, although our preference is to invert both jointly. Finally, the method, given appropriate computing resources, can be run efficiently on large collections of ambient noise cross correlations in an automated

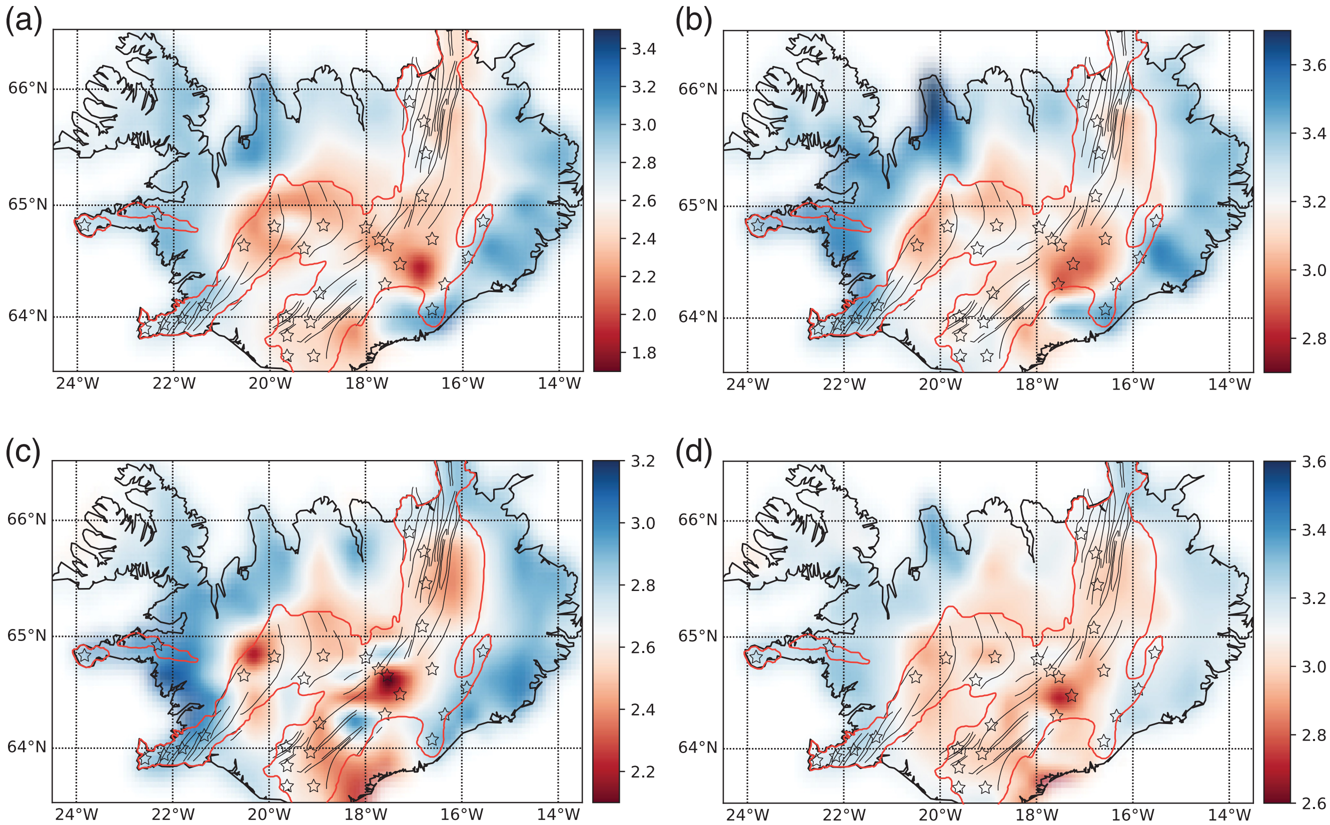

Figure 7. The results of tomographic inversions at the 6 s period of (a) Love-group velocity, (b) Love-phase velocity, (c) Rayleigh-group velocity, and (d) Rayleigh-phase velocity, with rifting zones obtained from Jóhannesson and Sæmundsson (1998) overplotted showing spatial correlation of slow velocity regions with known rifting zones and volcanism (volcanoes indicated with stars).

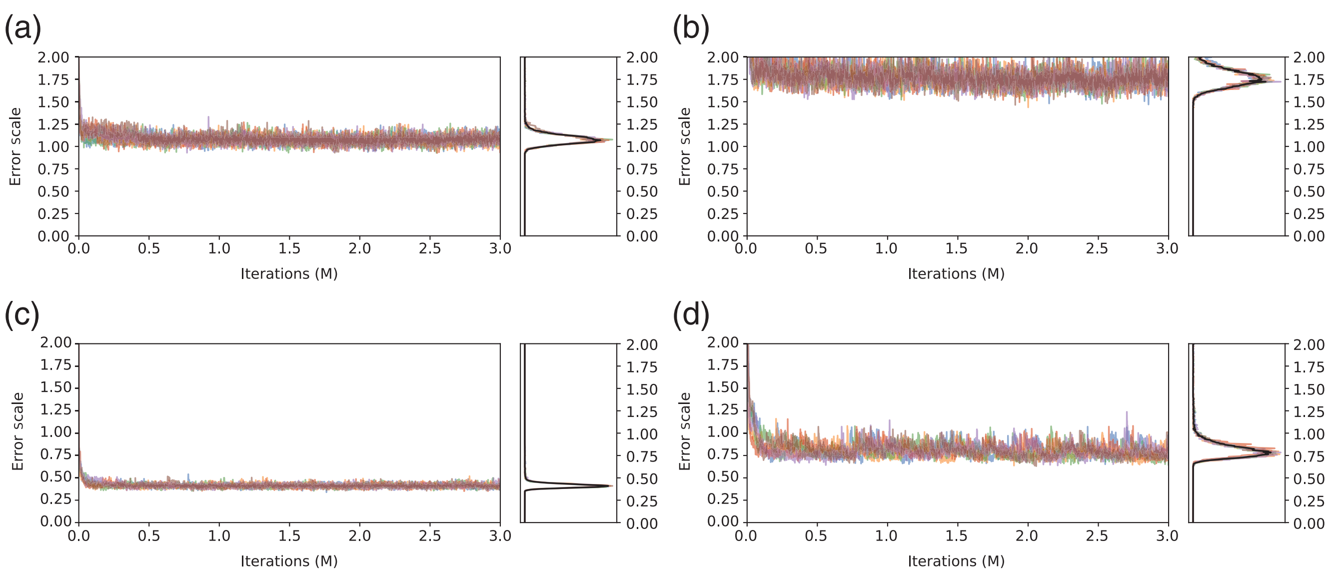

Figure 8. Convergence plots for the (a) Love-group velocity, (b) Love-phase velocity, (c) Rayleigh-group velocity, and (d) Rayleigh-phase velocity inversions performed at the 6 s period for the hierarchical scaling parameters. These values converge between 1.8 and 0.4 indicating that the linearized estimates of uncertainties are reasonably well estimated, that is, of the correct order of magnitude.

fashion, thereby allowing high-quality observations to be obtained for subsequent tomographic inversion for Earth structure.

Data and Resources

Raw seismogram data were downloaded from the facilities of the Incorporated Research Institutions for Seismology (IRIS) Data Services. Data were used from the HOTSPOT experiment available through http://dx.doi.org/10.7914/SN/XD_1996 and a single station from the Global Seismograph Network, BORG (http://www.fdsn.org/station_book/II/BORG/borg.html). All websites were last accessed on February 2018.

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